CS341 · COMPUTER GRAPHICS

Question Bank Practice

Practice on every midterm chapter

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01 Questions

A TRUE / FALSE · 1

1. In a random scan system, the picture is stored as a set of line-drawing commands in the display list. ( )
2. Random scan systems can easily display realistic shaded scenes. ( )
3. A pen plotter is a random scan device. ( )
4. A raster scan device scans the screen from top to bottom, one row at a time. ( )
5. Printers are raster scan devices. ( )
6. In an interlaced scan, the even lines are scanned first and then the odd lines. ( )
7. A monochrome 640 × 480 frame buffer needs 307,200 bytes. ( )
8. Lower-persistence phosphors require higher refresh rates to avoid flicker. ( )
9. High-persistence phosphors are useful for animation. ( )
10. An LCD is an emissive display. ( )

A TRUE / FALSE · 2

11. In a passive-matrix display, each pixel is controlled by its own transistor. ( )
12. For crisp images the dot pitch should be less than 0.31 mm. ( )
13. A point plus a vector is a point. ( )
14. 0.3P + 0.7R is a valid point. ( )
15. The cross product is defined for 2D vectors. ( )
16. Every affine combination is a convex combination. ( )
17. Point – Point = Vector. ( )
18. A unit vector has length 1. ( )
19. General displays are refreshed about 45 to 100 times per second. ( )
20. DDA uses only integer arithmetic. ( )

B MCQ · 1

1. Another name for a random scan display is:
   a) Raster b) Vector (calligraphic) c) Bit map d) Interlaced
2. In random scan systems, the picture definition is stored in the:
   a) Frame buffer b) Display list c) Bit map d) Video controller
3. Persistence is the time for emitted light to decay to:
   a) Half b) One-quarter c) One-tenth d) Zero
4. Which is NOT a hard-copy device?
   a) Film recorder b) Pen plotter c) Joystick d) Electrostatic printer
5. Screen clarity depends on:
   a) Resolution, dot pitch, refresh rate b) CPU, RAM, disk c) Color, size, weight d) Voltage only
6. The most important example of a nonemissive flat-panel display is:
   a) Plasma b) LED c) LCD d) Thin-film electroluminescent
7. Which flat-panel arrangement is brighter and sharper?
   a) Active-matrix b) Passive-matrix c) Both equal d) Neither

B MCQ · 2

8. If a = (3, 4), then |a| equals:
   a) 7 b) 5 c) 25 d) 12
9. The counterclockwise perpendicular of a = (2, 5) is:
   a) (5, −2) b) (−5, 2) c) (−2, −5) d) (2, −5)
10. In homogeneous form, a vector has a fourth coordinate equal to:
   a) 0 b) 1 c) −1 d) 2
11. The dot product of b = (1, 2) and c = (2, −1) is:
   a) 4 b) −3 c) 0 d) 1
12. Which of the following is NOT an input device?
   a) Mouse b) Light pen c) Pen plotter d) Joystick
13. A 10 × 5 inch screen with resolution 1920 × 1080 has an aspect ratio of:
   a) 0.89 b) 1.125 c) 1.78 d) 2
14. Bresenham's algorithm is preferred over DDA because it:
   a) Uses floating point b) Uses only integers c) Draws thicker lines d) Needs no starting point
15. Which OpenGL mode draws a closed polyline?
   a) GL_LINES b) GL_LINE_STRIP c) GL_LINE_LOOP d) GL_POINTS

C FILL IN THE BLANKS

1. Each row of a raster screen is called a _____.
2. A screen point is called a _____ (picture element).
3. The frequency at which a picture is redrawn is the _____.
4. The space between the centers of adjacent pixels is the _____.
5. In a raster system the picture is stored in the _____.
6. In a random scan system the picture is stored in the _____.
7. The vector from point P to point Q is _____.
8. Persistence is the time for the emitted light to decay to _____ of its original intensity.
9. Flat panels that convert electricity into light are _____; those that use optical effects are _____.
10. With GL_LINES and five vertices, vertex _____ is ignored.
11. The OpenGL utility library has the prefix _____ and the toolkit library _____.
12. The initial Bresenham line decision parameter is p0 = _____.

D SHORT ANSWER

1. Compare random scan and raster scan (storage, resolution, color).
2. Give two advantages and two disadvantages of random scan systems.
3. Name the parts of a CRT and the job of the deflection plates.
4. What is the difference between beam-penetration and shadow-mask color CRTs?
5. Compare active-matrix and passive-matrix displays.
6. Why do we use aspect ratio?
7. Compare DDA and Bresenham line algorithms.
8. What is the difference between absolute and relative coordinates?